Predicate Logic

16 March 2026

WHAT WE HAVE LEARNED SO FAR …

Our aim in this course

We are exploring the features of good arguments.

In a good argument, the truth of the premises should support the truth of that argument's conclusion.

What is an argument?

An argument is a collection of two things:

1. **A set of statements, which are known as *premises.***

2. **A single statement, which is known as a *conclusion*.**

Further, it is implied that there is a relationship between 1 and 2 such that the truth of 2 follows from 1.

It is this relationship, along with the larger property it reveals, which we are exploring.

This relationship is known as *entailment*, and in deductive arguments it is demonstrated by *validity*.

Validity

The principal objective of this course is to explore validity.

Validity is defined as follows:

An argument is valid **IF AND ONLY IF** it is **NOT** possible for the conclusion to be false when **ALL** the premises are true.

An interesting feature

With deductive arguments, the validity of the argument is a function of the underlying form of the argument.

For this reason we introduced a formal language that made it easier to analyze arguments.

The formal language allowed us to isolate and examine the underlying structure of arguments, making it easier to recognize patterns of validity.

This allowed us to become better at recognizing those structures in arguments constructed in natural languages.

How about an example?

Consider the following:

“If borders are secure, then terrorists cannot enter the country. If terrorists cannot enter the country, then acts of terrorism will be reduced. Therefore, if borders are secure, then acts of terrorism will be reduced.”

Recognize that it is an argument

- "If borders are secure, then terrorists cannot enter the country. If terrorists cannot enter the country, then acts of terrorism will be reduced. ***Therefore***, if borders are secure, then acts of terrorism will be reduced."

Extract its form

P1 - If borders are secure, then terrorists cannot enter the country.

P2 - If terrorists cannot enter the country, then acts of terrorism will be reduced.

C - If borders are secure, then acts of terrorism will be reduced."

Symbolize the sentences …

P1 - $(S \rightarrow {\sim}E)$

P2 - $({\sim}E \rightarrow R)$

C - $(S \rightarrow R)$

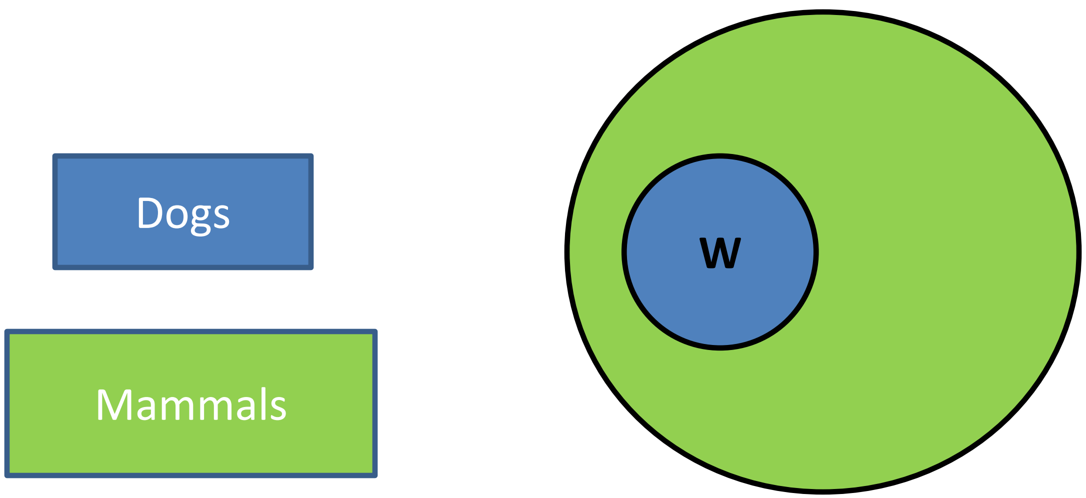
To test for Validity by Proof …

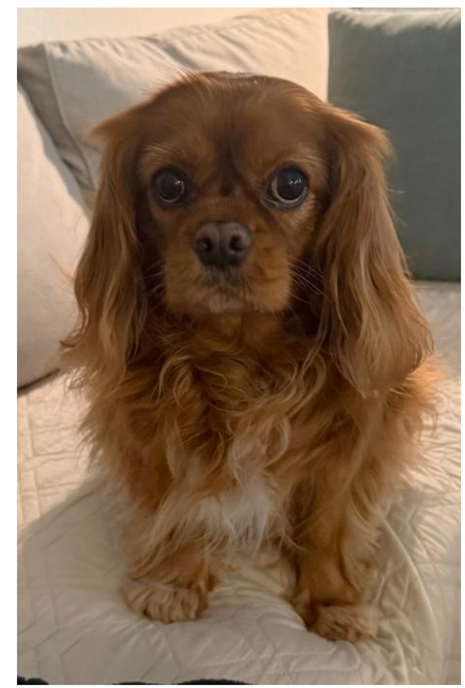
1	(1)	$(S \rightarrow \sim E)$	A
2	(2)	$(\sim E \rightarrow R)$	A
3	(3)	S	A (for $\rightarrow$I)
1,3	(4)	$\sim E$	1,2 $\rightarrow$E
1,2,3	(5)	R	2,4 $\rightarrow$E
1,2	(6)	$(S \rightarrow R)$	5 $\rightarrow$I (3)

But what about arguments like this:

All dogs are mammals, and Watson is a dog.
Therefore Watson is a mammal.

It appears valid, …





Watson

But representing its validity in SL fails

P1	-	D (All dogs are mammals.)
P2	-	W (Watson is a dog.)
C	-	M (Watson is a mammal.)

$D, W \vdash M$ is clearly an invalid sequent

What’s the problem?

- The problem is that some arguments that are valid cannot be represented as such with the tools we have explored so far.
- Why? Because the validity of these types of arguments stem from the internal structure of atomic sentences.
- From the perspective of Sentential Logic, that is hidden information.
- To represent these types of arguments we need a formal language that is robust enough to represent the internal structure of atoms.

What will that require?

- Digging inside atomic sentences to reveal their internal structure.

How do atomic sentences work?

In general, simple sentences work by identifying a *subject* and stating something about that subject. The “stating something about” is what is known as predication, hence why what we are now doing is known as predicate logic.

For example:

Sherlock	+	____ is a dog
Watson	+	____ is a dog
Reveille	+	____ is a dog
Something	+	____ is a dog
Everything	+	____ is a dog


AND NOW FOR SOMETHING
COMPLETELY DIFFERENT.

Terms in Traditional Logic

- In traditional, Aristotelian or Categorical logic, the different elements that combine to form sentences are known as ”terms.”

- A “term” is simply an idea that can be expressed in words, either spoken or written.

- There are four kinds:
  - Singular
  - Collective
  - Particular
  - Universal

The need for expanding our language

- The language of sentential logic is limited, e.g. it cannot adequately represent the following valid argument:

  Alice looked at Tom’s Facebook profile, therefore someone looked at Tom’s Facebook profile.

  Because this might be symbolically represented as: $P \vdash Q$ and that is an obviously invalid argument form.

What we need …

- Is a way of representing the internal structure of atomic sentences, i.e. what is hidden from us in the formal logic for Sentential logic that we learned in the earlier chapters.

To meet this need we will introduce:

- An expanded notion of what counts as an atomic sentence, and the following two new types of sentences:

- Universal sentences

- Existential sentences

Why do we need these?

- Because we need to represent the internals of how atomic sentences, from the perspective of Sentential Logic, work.

- Those sentences work by combining choosing *a subject* to talk about with *the something said about that subject*.

Vocabulary of Predicate Logic

- Sentence Letters
- Connectives
- Names
- Variables
- Predicate Letters
- Quantifiers
- Parentheses

(Our text mentions the Identity Symbol, but we won’t be using that in this class.)

Names

- To pick out specific subjects, we use names, e.g. Bob, Jane, etc., and represent them with lower-case letters from the start of the alphabet:

  $a, b, c, d, e, \ldots a_1, b_1, c_1, d_1, e_1, \ldots$

Variables

- In English we often have non-specific subjects, e.g. “someone,” “something,” “everyone,” etc. In order to help us represent these subjects, we need the help of variables, for which we will use lower-case letters from the latter half of the alphabet:

  $u, v, w, x, y, z, \ldots u_1, v_1, w_1, x_1, y_1, z_1, \ldots$

1-place Predicate Letter

- A **1-PLACE PREDICATE LETTER** is any symbol from the following list: A^1, B^1, C^1,…, A^1_1, B^1_1, C^1_1, A^1_2, B^1_2, C^1_2,…

  We use these to represent the portion of a sentence where a singular property is ascribed to a subject, e.g.

  ________ is a dog
  ________ is blue
  ________ is an Aggie

2-place Predicate letter

- A **2-PLACE PREDICATE LETTER** is any symbol from the following list: A^2, B^2, C^2,..., $A^2{}_1$, $B^2{}_1$, $C^2{}_1$, $A^2{}_2$, $B^2{}_2$, $C^2{}_2$,...

  In general, we use these to represent the portion of a sentence where a relational property is ascribed, e.g.

  ______ is next to _______
  ______ loves ________
  ______ hates ________

N-place Predicate Letter

- In general, an ***n*-PLACE PREDICATE LETTER** is any symbol from the list: A^n, B^n, C^n,..., A^n_1, B^n_1, C^n_1, A^n_2, B^n_2, C^n_2,...

  For more complex relations, we can create many-place predicate letters to represent things such as:

  _______ is between ________ and ________

Universal Quantifier

- When dealing with unspecific subjects, we usually need to reference quantity, e.g. “every,” for that we use the Universal Quantifier which we symbolize as:

$$\forall\alpha$$

(Where α is a variable)

Existential Quantifier

- When dealing with unspecific subjects, we usually need to reference quantity, e.g. “some,” for that we use the Existential Quantifier which we symbolize as:

$$\exists \alpha$$

(Where α is a variable)

Expression

- An **EXPRESSION OF PREDICATE LOGIC** is any sequence of items from the vocabulary of predicate logic.

Well-Formed Formulas

- A WELL-FORMED FORMULA of predicate logic is any expression in accordance with the following six rules:

(1) Sentence letters are wffs.

(2) An n-place predicate letter followed by n names is a wff.

(3) Negations, conjunctions, disjunctions, conditionals, and biconditionals of wffs are wffs.

(The formation rules of chapter 1 are subsumed by this clause.)

Universal WFF

(4) If ϕ is a WFF, then the result of replacing at least one occurrence of a name in ϕ by a new variable α (i.e., α not in ϕ) and prefixing $\forall\alpha$ is a WFF.

(Such WFFs are called "universally quantified," or just "universals.")

Existential WFF

(5) If ϕ is a WFF, then the result of replacing at least one occurrence of a name in ϕ by a new variable α (i.e., α not in ϕ) and prefixing $\exists\alpha$ is a WFF.

(Such WFFs are called "existentially quantified," or just "existentials.")

Nothing Else

- Nothing Else

Open Formula

- An **OPEN FORMULA** is the result of re-placing at least one occurrence of a name in a WFF by a new variable (one not already occurring in the WFF).

  (Open formulas are not WFFs, and hence never appear as sentences in proofs. The notion of an open formula is used to present the rules of proof for predicate logic.)

Scope

- The **SCOPE** of a quantifier in a given formula is the shortest open formula to the right of the quantifier. For example, in the following:

$$(\forall x Fx \ \& \ \exists y(Gy \ \& \ Hy))$$

The scope of the $\forall x$ is Fx

and the scope of the $\exists y$ is $(Gy \ \& \ Hy)$

- A quantifier whose scope contains another quantifier is said to have **WIDER SCOPE** than the second. The second is said to have **NARROWER SCOPE** than the first.

$$\forall x(Fx \rightarrow \exists yGy)$$

In this example the $\forall x$ has wider scope than the $\exists y$.

Universalization

- A **UNIVERSALIZATION** of a sentence with respect to a given name occurring in the sentence is obtained by the following two steps:

  1. Replace all occurrences of the name in the sentence by a variable α, where α does not already occur in the sentence.
  2. Prefix $\forall\alpha$ to the open formula resulting from step 1.

Existentialization

- An **EXISTENTIALIZATION** of a sentence with respect to a given name occurring in the sentence is obtained by the following two steps:
  1. Replace at least one occurrence of the name in the sentence by a variable α, where α does not already occur in the sentence.
  2. Prefix $\exists\alpha$ to the open formula resulting from step 1.

Instance

- An **INSTANCE** of a universally or existentially quantified sentence is the result of the following two steps:

  1. Remove the initial quantifier.
  2. In the open formula resulting from step 1, uniformly replace all occurrences of the unbound variable by a name.

  (This is called **INSTANTIATING** the sentence. The name is called the **INSTANTIAL NAME**.)